fn log_alpha_over_alpha_minus_one_lo

Michael Shoemate

March 16, 2026

Proves soundness of `fn log_alpha_over_alpha_minus_one_lo` in `mod.rs`.

1 Hoare Triple

Precondition

`alpha` > 1.

Pseudocode

```
1 def log_alpha_over_alpha_minus_one_lo(alpha: float):
2     inv_alpha_lo = 1.0.neg_inf_div(alpha)  #
3     neg_inv_alpha_hi = -inv_alpha_lo
4     ln_one_minus_inv_alpha_hi = neg_inv_alpha_hi.inf_ln_1p()  #
5     return -ln_one_minus_inv_alpha_hi
```

Postcondition

Theorem 1.1. For any setting of the input parameter `alpha` such that the given precondition holds, `log_alpha_over_alpha_minus_one_lo` either returns `Err(e)`, or returns `Ok(out)` where

$$\texttt{out} \leq \log\left(\frac{\texttt{alpha}}{\texttt{alpha} - 1}\right).$$

2 Proof

Assume the precondition is met.

Lemma 2.1. `log_alpha_over_alpha_minus_one_lo` only returns `Err(e)` if one of the fallible helper calls in the pseudocode returns `Err(e)`.

Proof. The only fallible operations are the calls on lines 2 and 4. □

Lemma 2.2. If the outcome of `log_alpha_over_alpha_minus_one_lo` is `Ok(out)`, then

$$\texttt{out} \leq \log\left(\frac{\texttt{alpha}}{\texttt{alpha} - 1}\right).$$

Proof. By the proof definition of `neg_inf_div`,

$$\texttt{inv_alpha_lo} \leq \frac{1}{\texttt{alpha}}.$$

Multiplying by -1 reverses the inequality:

$$\texttt{neg_inv_alpha_hi} = -\texttt{inv_alpha_lo} \geq -\frac{1}{\texttt{alpha}}.$$

Since $\texttt{alpha} > 1$, both arguments lie in $(-1, 0)$, so by the proof definition of `inf_ln_1p`,

$$\texttt{ln_one_minus_inv_alpha_hi} \geq \ln\left(1 - \frac{1}{\texttt{alpha}}\right).$$

Multiplying by -1 gives

$$\texttt{out} = -\texttt{ln_one_minus_inv_alpha_hi} \leq -\ln\left(1 - \frac{1}{\texttt{alpha}}\right) = \log\left(\frac{\texttt{alpha}}{\texttt{alpha} - 1}\right).$$

□

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.2. □